

Multistage germanium-rich sphalerite and wurtzite mineralization from the Qingshan carbonate-hosted Zn-Pb deposit, SW China

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ABSTRACT

As a critical raw material for green technologies, germanium (Ge) is frequently concentrated in sphalerite (cubic ZnS). In nature, ZnS also occurs as its hexagonal polymorph, known as wurtzite; however, the distribution and incorporation of Ge in wurtzite still remains poorly understood. Using laser ablation–inductively coupled plasma–mass spectrometry analysis, this study investigates Ge distribution in sphalerite and wurtzite from three mineralization stages at the Qingshan carbonate-hosted Zn-Pb deposit, Southwest China. Stage I sphalerite (Sp-1) is enriched in Ge (109–672 ppm), whereas Stage II sphalerite (Sp-2) displays Ge-rich (dark Sp-2a; 174–899 ppm) and Ge-poor (light Sp-2b; ≤14.8 ppm) domains. Colloform wurtzite precipitated in Stage III (Wz-3) contains 58.0–722 ppm Ge, followed by euhedral sphalerite (Sp-3) with the highest Ge contents (467–1393 ppm). Combining transmission electron microscopy with trace element content correlations reveals that Ge is a structurally bound element in both sphalerite and wurtzite, incorporated via direct substitution mechanisms such as $\text{Ge}^{2+} \leftrightarrow \text{Zn}^{2+}$ and $\text{Ge}^{4+} + \text{vacancy} \leftrightarrow 2\text{Zn}^{2+}$. *Ab initio* calculations demonstrate no correlation between ZnS crystal structures (cubic/hexagonal) and Ge incorporation levels. The strongly positive $\delta^{34}\text{S}$ values in Wz-3 (+17.2‰ to +19.2‰) suggest that the colloform texture is independent of low temperatures (<100 °C) and bacterial sulfate reduction. Increasing sulfur fugacity triggered wurtzite-to-sphalerite phase transition (Wz-3→Sp-3) and may have enhanced the incorporation of Ge in Sp-3. We propose

a Ge enrichment process involving coupled dissolution-reprecipitation. This study highlights the importance of coupling micro- to nano-scale textural and chemical observations in ZnS minerals for understanding Ge enrichment.

INTRODUCTION

Germanium (Ge) is considered a critical metal due to its high economic importance and the risk of supply scarcity (Gulley et al., 2018; European Commission, 2023; U.S. Geological Survey, 2024). It is an essential raw material for infrared optical devices and is widely used in communications and renewable energy technologies. Despite its low background concentration in the continental crust (~1.5 ppm; Smith and Huyck, 1999), Ge can be enriched in a wide variety of geological settings because it has different geochemical behaviors, including siderophile, lithophile, chalcophile, and organophile properties (Bernstein, 1985). Presently, Ge is economically recovered as a by-product from zinc sulfide ores and coals (Höll et al., 2007; Frenzel et al., 2014, 2017).

Sphalerite (cubic ZnS) is one of the most common hosts for Ge, in which Ge may occur as solid solutions and/or as micro- to nanoscale Ge minerals (e.g., argutite $[\text{GeO}_2]$, brunogeierite $[\text{GeFe}_2\text{O}_4]$, and briartite $[\text{Cu}_2(\text{Zn},\text{Fe})\text{GeS}_4]$; Horn et al., 2019; Cugerone et al., 2018, 2021; Fougereuse et al., 2023; Niu et al., 2024). Germanium distribution in sphalerite is heterogeneous, related to crystallographic controls (e.g., sector zoning and rhythmic bands; Belisont et al., 2014; Wang et al., 2023b), mineral morphology (e.g., colloform and euhedral; Luo et al., 2022), and post-crystallization remobilization (Cugerone et al., 2020). The complex physicochemical processes during the crystallization of sphalerite are responsible for these observations (Frenzel and Thiele, 2024). For instance, colloform sphalerite is favorable for Ge enrichment due to its specific formation conditions,

which include low temperature, low pH, and a rapid growth rate (Luo et al., 2022). Therefore, reconstructing the evolution of the physicochemical conditions in the precipitating fluid during sphalerite crystal growth via textural and chemical observations is crucial for understanding Ge enrichment.

Zinc sulfide also exhibits a hexagonal structure, known as wurtzite. Although the phase relations between wurtzite and sphalerite have been debated for many years, there is direct evidence of the coexistence of both phases in sulfide ores (e.g., Ciobanu et al., 2011; Bonnet et al., 2016; Torró et al., 2023). Wurtzite is apparently much rarer in nature than sphalerite but may frequently be overlooked because ZnS is usually assumed to be sphalerite (Pring et al., 2020). In Zn mines of Tennessee, USA, Bonnet et al. (2016) reported that hexagonal wurtzite is more enriched in Cu, Ge, and Ga compared to cubic sphalerite, and emphasized a possible relationship between the trace element contents and ZnS crystal structures. It is important to investigate the controlling role of the two crystal structures in Ge enrichment. However, the distribution and incorporation of trace elements such as Ge in these hexagonal crystals have not received much attention.

Carbonate-hosted Zn-Pb deposits are of interest due to their potential for economically recoverable Ge (Höll et al., 2007). In these deposits, sphalerite can contain up to a few thousands of ppm Ge (e.g., Horn et al., 2019; Torró et al., 2023; Sun et al., 2024). Here, we investigated the micro- to nano-scale texture and chemical composition of multistage sphalerite and wurtzite from the Qingshan carbonate-hosted Zn-Pb deposit, Southwest China. This study focused on the incorporation of Ge into sphalerite and wurtzite, as well as the influence of these two crystal structures on Ge enrichment. Additionally, detailed textural and chemical observations allow us to decipher the physicochemical processes controlling variable Ge concentrations.

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GEOLOGICAL SETTING

Regional Geology

The Sichuan-Yunnan-Guizhou region, located on the southwestern margin of the Yangtze Block (Fig. 1A), is one of the most important Zn and Pb mining districts in China (Zhang et al., 2015). The metamorphic basement of the region mainly consists of late Archaean metamorphic rocks and younger, weakly metamorphosed Mesoproterozoic to Neoproterozoic sedimentary successions including the Dongchuan, Huili, and Kunyang groups (Fig. 1B; Hu et al., 2017). The overlying sedimentary cover comprises the Neoproterozoic Ediacaran to Mesozoic Triassic submarine

sedimentary sequences along a passive margin (Fig. 1B; Xu et al., 2020), which are represented by a package of carbonate, siliciclastic, shale, and evaporitic rocks.

Structurally, the Sichuan-Yunnan-Guizhou region is dominated by the NW-trending Ziyun-Yadu, NE-trending Mile-Shizong, and N-trending Anninghe fault belts (Fig. 1B). The region experienced major periods of magmatism in the Late Permian (ca. 260–257 Ma; Zhong et al., 2020; Huang et al., 2022), resulting in the widely distributed Emeishan flood basalts that could have contributed to Zn-Pb mineralization (Fig. 1B; Xu et al., 2014; Zhou et al., 2018). Less voluminous intrusive activity in the region occurred as small diabase dikes near some Zn-Pb

deposits (Fig. 1B). Zircon from these basic intrusive rocks yielded a U-Pb age of 268.3 ± 7.4 Ma (Qi et al., 2016).

More than 450 Zn-Pb deposits, associated with critical metals such as Ga, Ge, and In (Meng et al., 2024), have been discovered in the Sichuan-Yunnan-Guizhou region (Fig. 1B; Han et al., 2023). Most of these Zn-Pb deposits are hosted in Ediacaran to Permian carbonate rocks, with a minority occurring in low-grade metamorphic schist of the Mesoproterozoic basement (Wang et al., 2023a). The Sichuan-Yunnan-Guizhou region contains Zn-Pb deposits that formed at ca. 458–366 Ma (Xiong et al., 2018; Liu et al., 2018; Wang et al., 2019) and 230–200 Ma (Hu et al., 2017, and references therein). These two

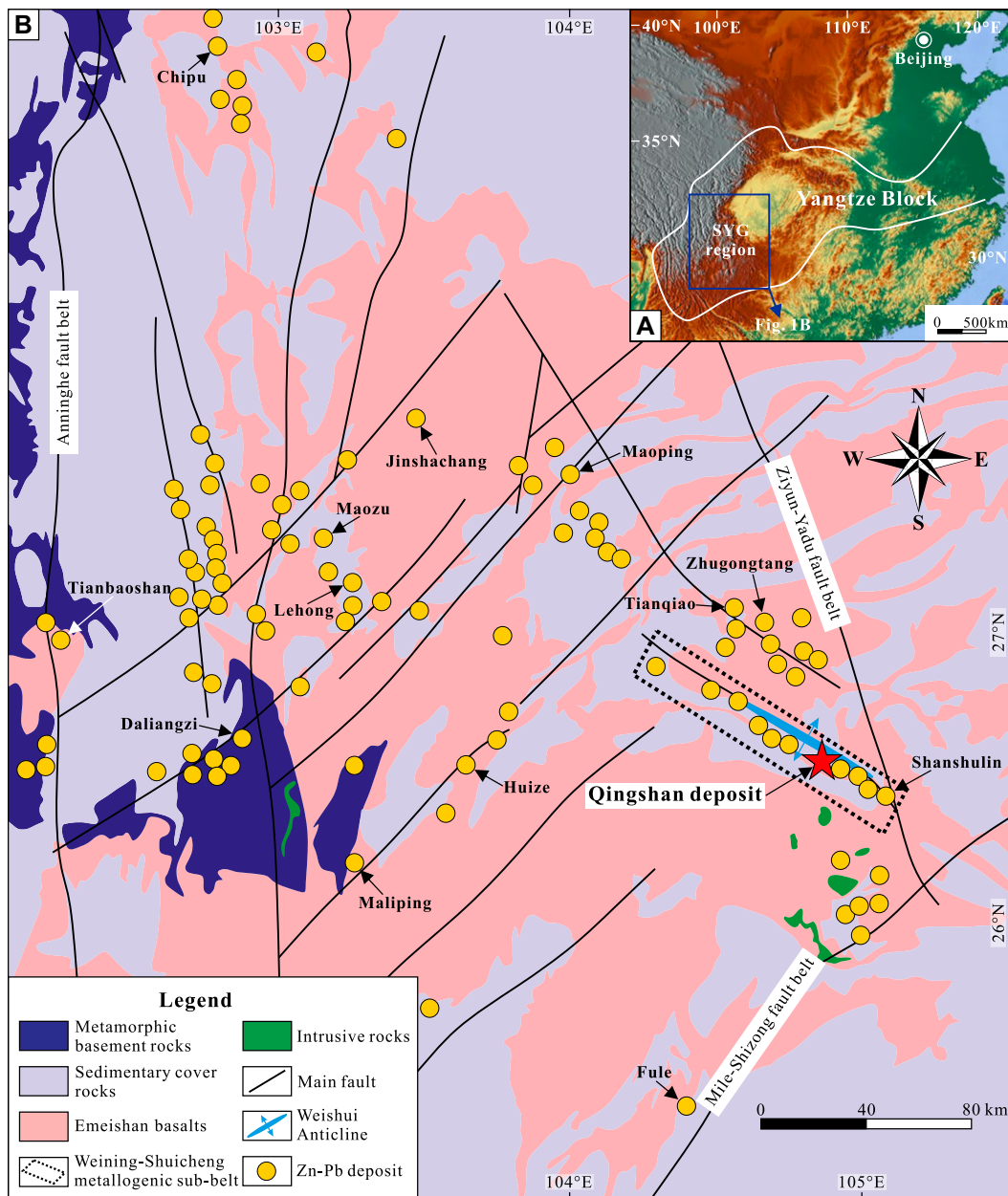


Figure 1. (A) Sketch map showing the location of the Sichuan-Yunnan-Guizhou (SYG) region within the Yangtze Block, South China. (B) Simplified geological map of the SYG region and location of main Zn-Pb deposits (modified after Luo et al., 2019; Wei et al., 2024). The Qingshan deposit is highlighted with a red star.

age clusters coincide with the late Caledonian to early Variscan collisional compressional event and the late Indosinian to early Yanshanian post-collisional extensional event, respectively (Li et al., 2010; Wang et al., 2013).

Ore Deposit Geology

The Qingshan deposit is located in the Wein-ing-Shuicheng metallogenic sub-belt, a terrain dominated by the Weishui Anticline (Fig. 1B; Tang et al., 2019). The deposit has been mined since 1993, yielding >1.0 Mt of Zn-Pb ore at average grades of 32.6% Zn and 7.7% Pb, and exploration efforts are ongoing.

The major economic mineralization at Qingshan is hosted in micrite limestone of the Upper Carboniferous Maping Formation (~150 m thick), which is unconformably overlain by the Middle Permian Liangshan Formation that consists of organic-rich sandstone and shale (Fig. 2A). Structurally, it is localized between two normal faults (F_1 and F_2), which both trend WNW and have a steep dip (70° – 80°) toward the SSW (Figs. 2A and 2B). This fault system is interpreted as a major structural control for the ore mineralization, acting as a conduit for metal-rich fluids to migrate into the overlying and adjacent favorable host facies (e.g., Song et al., 2024). Diabase dikes are exposed on the western side of the ore district (Fig. 2A), but no clear relationship with ore mineralization was noticed in the field.

At Qingshan, 12 major orebodies are documented and classified into two types according to their dip angle. The steeply dipping orebodies with a dip of 70° – 80° are stratabound, whereas the gently dipping orebodies with a dip of $\sim 20^\circ$ crosscut the bedding of the limestone (Fig. 2B). Mineral assemblage within the sulfide ore is relatively simple. The sulfide minerals include sphalerite, galena, pyrite, and a small amount of marcasite, while calcite is the predominant gangue mineral (Wu et al., 2023). The Zn-Pb mineralization is spatially linked to Fe-Mn carbonate alteration (Zhou et al., 2013).

SAMPLES AND ANALYTICAL METHODS

Our samples were collected from the underground galleries (nos. 1580, 1643, 1691, and 1755 levels) in the Qingshan district. A total of 36 thick polished sections ($\sim 150\ \mu\text{m}$ in thickness) were prepared for optical microscopy using reflected and transmitted plane-polarized light. Areas of interest were subsequently marked for further studies.

Scanning Electron Microscopy

Scanning electron microscopy was carried out at Guangzhou Tuoyan Analytical Technology Co., Ltd., Guangzhou, China, using a Tescan Mira 3 field emission scanning electron microscope. The backscattered electron images were

acquired under an operating voltage of 25 kV and a working distance of $\sim 15.0\ \text{mm}$.

Transmission Electron Microscopy

Two thin foil samples were prepared using a focused ion beam–scanning electron microscope system (Tescan Amber) at the Guangzhou Institute of Geochemistry, Chinese Academy of Sciences (GIGCAS). The focused ion beam was operated at an accelerating voltage of 30 kV to produce thin foils of $\sim 100\ \text{nm}$ in thickness through a series of processes, including Pt coating, trench cutting, and final milling. Detailed procedures followed those described by Ciobanu et al. (2011). Transmission electron microscopy (TEM) analyses were conducted using a Thermo Fisher Scientific FEI Talos F200S microscope operated at 200 kV at the GIGCAS. Various techniques, including scanning TEM (STEM), high-resolution TEM (HRTEM), selected area electron diffraction, and energy dispersive spectroscopy, were employed for morphological, structural, and compositional investigations.

Trace Element Spot and Mapping Analyses

In situ trace element contents were determined by laser ablation–inductively coupled plasma–mass spectrometry (LA-ICP-MS) using an NWR 193 nm ArF excimer laser system attached to a Thermo Fisher Scientific iCAP RQ ICP-MS at Guangzhou Tuoyan Analytical

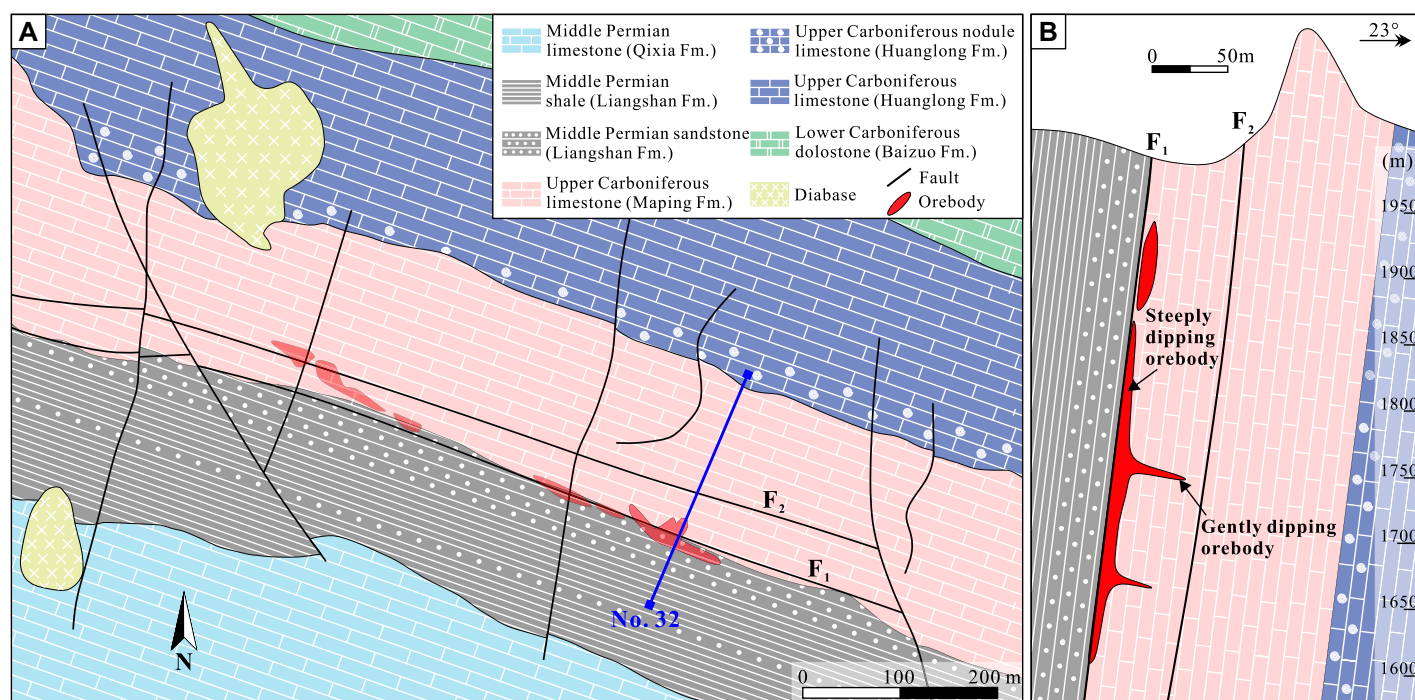


Figure 2. (A) Geology map of the Qingshan mining district. (B) Cross section of No. 32 exploration line.

Technology Co. Ltd., Guangzhou, China. The laser system was set to ablate a 35 μm -diameter spot using a fluence of 3.5 J/cm^2 at a repetition rate of 6 Hz. Helium is used as the carrier gas and argon as the makeup gas to adjust sensitivity during the laser ablation process. The following isotopes were monitored during each measurement (40 s background, 45 s ablation): ^{55}Mn , ^{57}Fe , ^{65}Cu , ^{71}Ga , ^{74}Ge , ^{75}As , ^{107}Ag , ^{111}Cd , ^{115}In , ^{118}Sn , ^{121}Sb , ^{205}Tl , and ^{208}Pb . External calibration without internal standards was performed using glass standards U.S. National Institute of Standards and Technology (NIST) SRM610 and MASS-1 (Liu et al., 2008), and NIST SRM612 and STDGL3-52 were used as monitoring samples. Offline data processing was performed by Iolite software (Paton et al., 2011).

Trace element mapping was conducted using the same LA-ICP-MS instrument at Guangzhou Tuoyan Analytical Technology Co. Ltd., Guangzhou, China. During the mapping, a laser energy density of 5 J/cm^2 and a laser repetition rate of 15 Hz were utilized, with a spot diameter of 10 μm and a scan speed of 40 $\mu\text{m}/\text{s}$. The reference material NIST SRM610 was analyzed as an external standard. Data reduction was conducted using Iolite software by Paton et al. (2011).

Sulfur Isotope Analysis

In situ sulfur isotope analysis was conducted with a Neptune Plus multi collector–ICP-MS (Thermo Fisher Scientific) equipped with a Geolas HD excimer 193 nm ArF laser system at Wuhan Sample Solution Analytical Technology Co. Ltd., Wuhan, China. Helium gas (650 mL/min) was used as a carrier to transport the ablated materials into the plasma. Laser ablation was performed with a spot diameter of 44 μm , a pulse frequency of 2 Hz, and a fluence of 6 J/cm^2 . Each acquisition incorporated 30 s of background measurement followed by 60 s of sample ablation. Sphalerite standard NBS-123 was chosen as a reference material for correcting the data. The in-house reference material sphalerite LI-SP-01 was analyzed repeatedly as an unknown sample to control the data quality. All $\delta^{34}\text{S}$ values were reported relative to Vienna Canyon Diablo Troilite.

Density Functional Theory Calculations

Density functional theory calculations were conducted with CASTEP code (Clark et al., 2005). In this study, the generalized gradient approximation–Perdew–Burke–Ernzerhof exchange–correlation functional (Perdew et al., 1992) was applied together with ultrasoft pseudopotentials (Vanderbilt, 1990). Geometry optimizations were achieved with convergence thresholds of 0.001 Å for displacement, 0.03 eV/Å for acting force, 0.05 GPa for internal stress, and 1.0×10^{-6} eV/atom for self-consistent field tolerance. Through convergence tests, the energy cutoff was 500 eV for both sphalerite and wurtzite, and the Monkhorst-Pack k point meshes of sphalerite and wurtzite were $4 \times 4 \times 4$ and $6 \times 6 \times 4$, respectively (Tables S1 and S2 in the Supplemental Material¹).

Geometry optimizations of sphalerite were performed using a $2 \times 2 \times 2$ supercell containing 64 atoms (32 Zn and 32 S), based on the crystal structure data (space group: $F\bar{4}3m$; cell parameters: $a = b = c = 5.406$ Å; Skin-

ner, 1961; Job 1 in Table 1). Geometry optimizations of wurtzite-2H were conducted using a $3 \times 3 \times 2$ supercell containing 72 atoms (36 Zn and 36 S), based on the crystal structure data (space group: $P6_3mc$; cell parameters: $a = b = 3.817$ Å, $c = 6.256$ Å; Cui et al., 1992; Job 2 in Table 1). The deviations between calculated and experimental lattice parameters are all $<2\%$ (Table S3), indicating that the chosen computational method is reliable for optimizing the sphalerite and wurtzite crystal structures.

Two geometry optimizations were conducted to separately characterize the isomorphic substitution of Zn^{2+} by Ge^{2+} in sphalerite and wurtzite, respectively (Jobs 3 and 4 in Table 1). Two geometry optimizations of Ge^{4+} substitution in sphalerite were conducted by replacing two Zn^{2+} with one Ge^{4+} and one vacancy at different relative positions (Jobs 5 and 6 in Table 1). Two geometry optimizations were conducted to investigate the substitution of Ge^{4+} in wurtzite by replacing two Zn^{2+} with one Ge^{4+} and one vacancy at different relative positions (Jobs 7 and 8 in Table 1).

RESULTS

Petrography and Mineralogy

The Qingshan deposit contains two ore types: massive and brecciated (Figs. 3A–3E). Massive ore is generally predominant; however, brecciated ore is more than twice as abundant as massive ore in some orebody outcrops (e.g., 1691 level in underground galleries). Massive ore (>90 vol% sulfides) is composed mainly of pyrite, sphalerite, and galena, with minor calcite (Figs. 3C and 3D). The main textural feature of massive ore is anhedral sphalerite (Sp-1) and galena cementing fractured aggregates of pyrite (Figs. 4A and 4B). Under transmitted plane-polarized light, Sp-1 typically displays reddish-brown to yellowish-brown coloration (Fig. 4C).

Brecciated ore is characterized by centimeter- to decimeter-sized sulfide clasts (with colloform morphology) cemented by white calcite (Fig. 3E). These sulfide clasts typically have a core–rim texture (Fig. 4D). In the core, porous sphalerite (Sp-2) occurs as coarse-grained,

TABLE 1. LIST OF GEOMETRY OPTIMIZATIONS FROM DENSITY FUNCTIONAL THEORY CALCULATIONS

Job ID	Optimized geometry	Atoms	Note
1	Standard sphalerite	32 Zn, 32 S	
2	Standard wurtzite	36 Zn, 36 S	
3	Ge^{2+} substitution in sphalerite	1 Ge, 31 Zn, 32 S	
4	Ge^{2+} substitution in wurtzite	1 Ge, 35 Zn, 36 S	
5	Ge^{4+} substitution in sphalerite (1)	1 Ge, 30 Zn, 32 S	Ge^{4+} is adjacent to a vacancy; Ge–vacancy distance is 3.81 Å
6	Ge^{4+} substitution in sphalerite (2)	1 Ge, 30 Zn, 32 S	Ge^{4+} is distant to a vacancy; Ge–vacancy distance is 6.59 Å
7	Ge^{4+} substitution in wurtzite (1)	1 Ge, 34 Zn, 36 S	Ge^{4+} is adjacent to a vacancy; Ge–vacancy distance is 3.81 Å
8	Ge^{4+} substitution in wurtzite (2)	1 Ge, 34 Zn, 36 S	Ge^{4+} is distant to a vacancy; Ge–vacancy distance is 7.38 Å

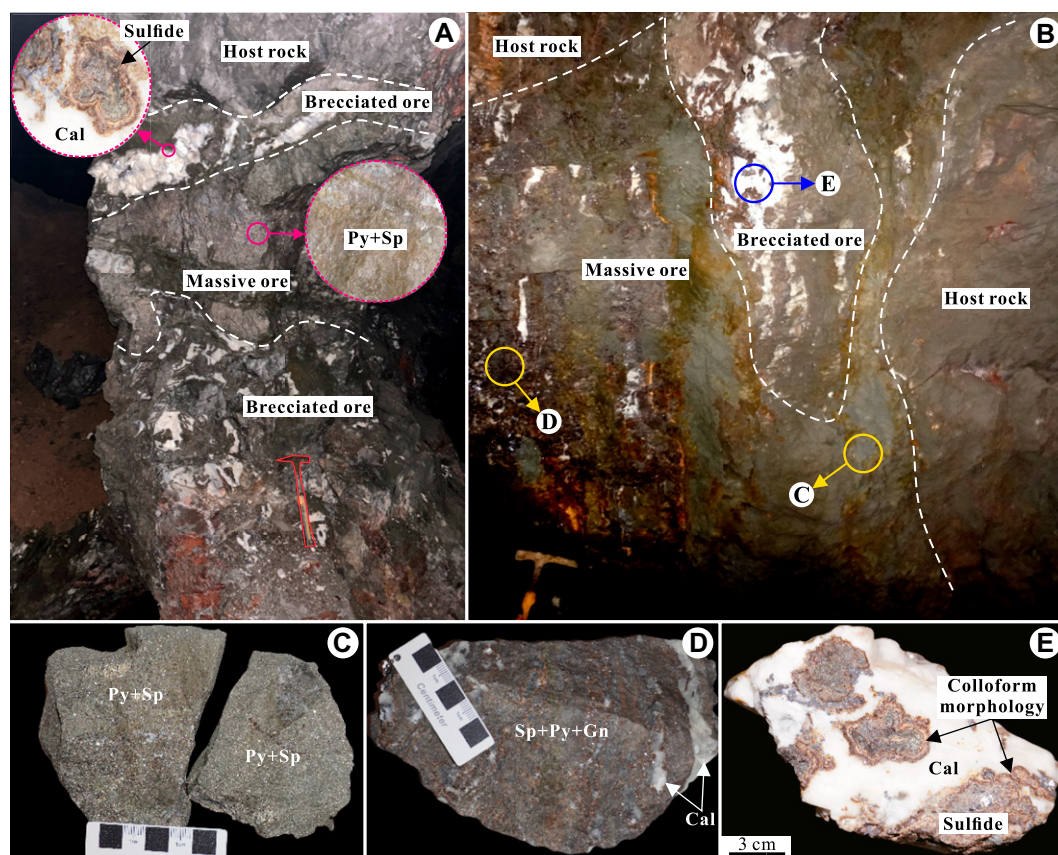


Figure 3. (A) Outcrop of gently pitching orebody with massive and brecciated mineralization. (B) Outcrop of steeply pitching orebody, showing massive mineralization interspersed by brecciated mineralization. (C) Massive ore dominated by pyrite (Py) and sphalerite (Sp). (D) Massive ore containing sphalerite, pyrite, and galena (Gn) with minor calcite (Cal). (E) Hand specimen of brecciated ore, showing sulfide clasts with colloform morphology.

subhedral crystals and is associated with fragmented pyrite of various sizes (Fig. 4D). Under transmitted plane-polarized light, Sp-2 shows two distinct color zones that include dull-brown to yellow (Sp-2a) and white (Sp-2b) crystals (Fig. 4E). Commonly, Sp-2b is late and occurs around Sp-2a (Fig. 4E). Backscattered electron imaging reveals a brighter gray contrast for Sp-2b relative to Sp-2a (Fig. 4F). In the rim, colloform ZnS is overgrown by euhedral sphalerite (Sp-3), indicating that Sp-3 precipitation directly followed colloform ZnS formation (Figs. 4D and 4E). The thicknesses of colloform ZnS layers generally vary from 2 mm to 10 mm in all samples. Under transmitted plane-polarized light, colloform ZnS shows oscillatory color zoning with black to light-yellow color zones (Fig. 4E). The backscattered electron image further reveals multiple, finer-scale zonings in colloform ZnS (Fig. 4G). Locally, colloform ZnS contains fine-grained inclusions of galena (Figs. 4D and 4G). Euhedral Sp-3 crystals, oriented perpendicular to colloform ZnS layers, are mostly brown to yellow in transmitted plane-polarized light (Fig. 4E).

TEM was used to characterize the crystal structures of colloform ZnS and euhedral Sp-3 in focused ion beam foils. The HRTEM images of colloform ZnS reveal a d-spacing of 0.625 nm

for the (0001) plane, which corresponds to the ideal d-spacing for the same plane in the hexagonal 2H structure polytype of wurtzite (Figs. 5A and 5B; Table 2). The fast Fourier transform image further confirms the crystal structure of colloform ZnS by showing a hexagonal phase with a $[1\bar{2}10]$ zone axis (Fig. 5C). By comparison, Sp-3 exhibits d-spacings of 0.191 nm and 0.270 nm in HRTEM images, which match the ideal d-spacings of (220) and (020) planes in sphalerite, respectively (Figs. 5D and 5E; Table 2). Furthermore, the fast Fourier transform image shows a face-centered cubic phase with a $[001]$ zone axis (Fig. 5F). Notably, wurtzite (Wz-3) is characterized by abundant stacking fault defects, as shown by HRTEM images (Figs. 5A and 5B).

The high-angle annular dark field STEM (HAADF-STEM) image performed on Wz-3 reveals the presence of nanoparticles with sizes ranging from 50 nm to 100 nm (Fig. 5G). Corresponding energy dispersive spectroscopy maps identify these nanoparticles as galena inclusions within Wz-3 (Figs. 5H and S1). In contrast, no inclusions are observed in Sp-3 from HAADF-STEM images (Figs. 4I and S2). Our petrological and mineralogical investigations establish a simple paragenetic sequence for the Qingshan deposit (Fig. 6).

Chemical Composition and Distribution

Of the analyzed elements, Sb, As, and Pb occasionally gave spiky LA-ICP-MS transient signals, which are possibly indicative of micro-inclusions (e.g., galena). Therefore, based on systematic flat time-integrated LA-ICP-MS signals, we selectively reported data from ablation pits considered to represent solid solutions, thereby excluding analyses contaminated by such micro-inclusions. The results of LA-ICP-MS spot analyses for Sp-1 ($n = 19$), Sp-2a ($n = 23$), Sp-2b ($n = 23$), Wz-3 ($n = 41$), and Sp-3 ($n = 34$) are summarized in Table 3 and illustrated in Figure 7. The complete LA-ICP-MS dataset (including detection limits) is provided in Table S6.

Sp-3 contains the highest Ge contents, ranging from 467 ppm to 1393 ppm (median 827 ppm; standard deviation 212 ppm). Sp-1, Sp-2a, and Wz-3 are also rich in Ge, with 109–672 ppm (median 399 ppm; standard deviation 165 ppm), 174–899 ppm (median 333 ppm; standard deviation 211 ppm), and 58.0–722 ppm (median 211 ppm; standard deviation 157 ppm), respectively. In contrast, Sp-2b is Ge-poor, with most contents below 14.8 ppm (median 4.53 ppm; standard deviation 3.56 ppm).

For all samples, the contents of Fe and Cd range from 1720 ppm to 66,736 ppm and 309–

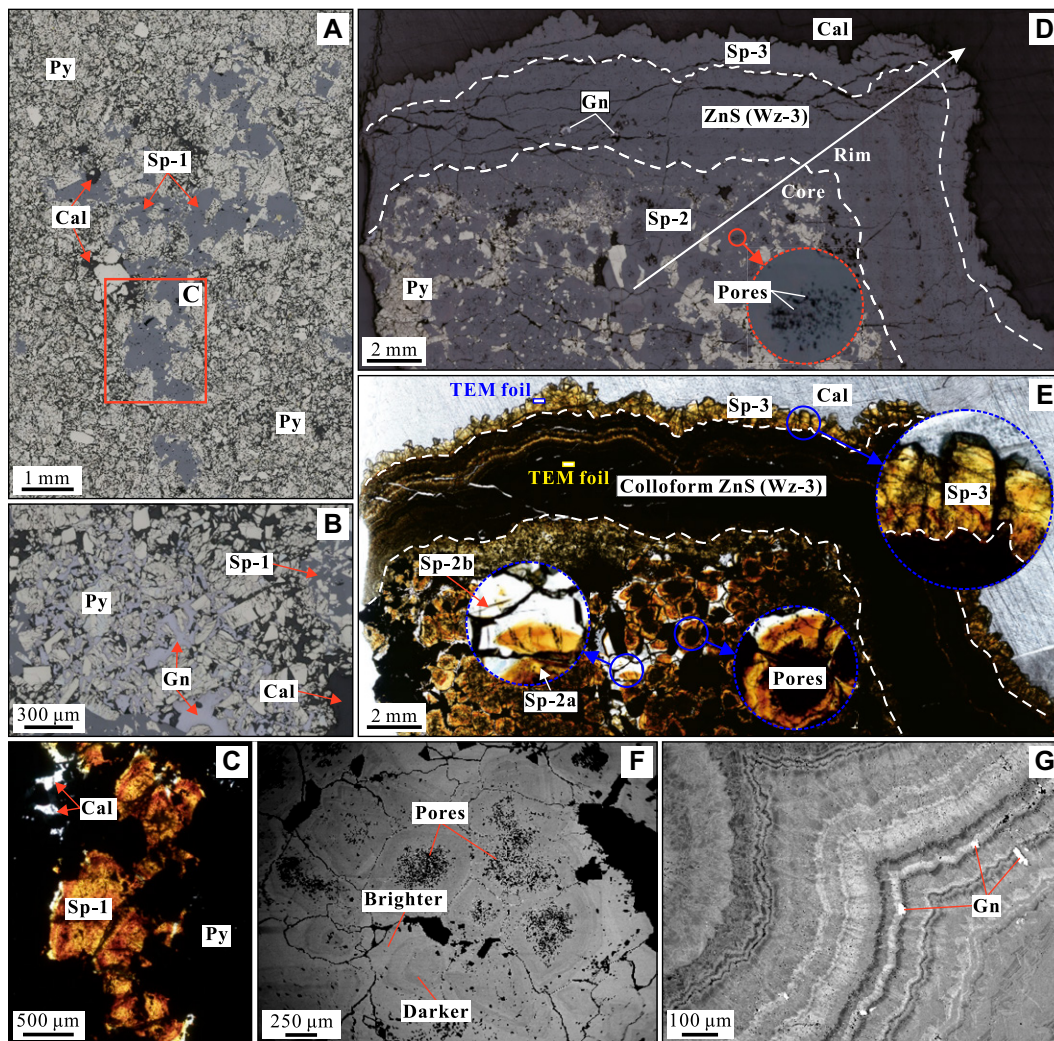


Figure 4. (A, B) Fractured pyrite cemented by Sp-1 and galena (reflected plane-polarized light). (C) Anhedrous Sp-1 with reddish-brown to yellowish-brown colors (transmitted plane-polarized light). (D) Representative photomicrograph displaying a core-rim texture at the sulfide clast (reflected plane-polarized light). (E) Dull-brown to yellow Sp-2a and white Sp-2b occurring in the core, followed colloform Wz-3 overgrown by euhedral Sp-3 (transmitted plane-polarized light). (F) Backscattered electron (BSE) image of Sp-2 with brighter (Sp-2a) and darker (Sp-2b) domains. Note porous textures. (G) Oscillatory zoned Wz-3 containing micron-sized Gn inclusions (BSE image). Sp—sphalerite; Py—pyrite; Gn—galena; Wz—wurtzite; Cal—calcite.

13,888 ppm, respectively. Sp-1 has the highest Fe contents (29,646–66,736 ppm; median 41,130 ppm), whereas the lowest Fe contents (1720–11,286 ppm; median 2740 ppm) are observed in Sp-2b. Cadmium contents are broadly comparable in Sp-2b (median 3558 ppm), Wz-3 (median 4576 ppm), and Sp-3 (median 4335 ppm).

The highest Mn contents (14.7–431 ppm; median 86.3 ppm) and Ga contents (0.63–77.4 ppm; median 20.3 ppm) are recorded in Sp-2a and Sp-2b, respectively. The contents of Cu (18.4–1400 ppm; median 542 ppm), As (27.6–3093 ppm; median 428 ppm), Ag (13.9–2034 ppm; median 132 ppm), Sb (7.24–2042 ppm; median 113 ppm), and Pb (203–8892 ppm; median 1414 ppm) in Wz-3 may reach high levels. The contents of Sn and Tl remain below 90.0 ppm. Indium contents are negligible with contents below 0.48 ppm.

The spatial distributions of Mn, Fe, Cu, Ga, Ge, Ag, Cd, and Pb are shown for Sp-2a and

Sp-2b in Figure 8 as well as Wz-3 and Sp-3 in Figure 9. Sp-2a is relatively rich in Mn, Fe, and Ge, and depleted in Ga and Cd compared to Sp-2b (Fig. 8). In Sp-2a, Ge contents are positively correlated with Fe and negatively correlated with Cd (Fig. 8). A significant discrepancy in chemical distributions between Wz-3 and Sp-3 is revealed. Wz-3 is relatively rich in Cu, Ag, Cd, and Pb, and depleted in Fe and Ge compared to Sp-3 (Fig. 9). The compositional map shows Ge is positively correlated with Fe in Wz-3, and with Fe, Cd, and Pb in Sp-3 (Fig. 9).

Sulfur Isotope Compositions

The sulfur isotope compositions of Wz-3 and Sp-3 are presented in Table S7. The result displays a range of relatively homogeneous $\delta^{34}\text{S}$ values from +16.6‰ to +19.2‰ (mean = +17.8‰; $n = 22$). Wz-3 has $\delta^{34}\text{S}$ values between +17.2‰ and +19.2‰ (mean = +18.2‰; $n = 14$), and

Sp-3 shows $\delta^{34}\text{S}$ values ranging from +16.6‰ to +17.6‰ (mean = +17.0‰; $n = 8$).

Formation Energies and Bond Distances

The formation energies of different systems in Jobs 3–8 (Table 1) were calculated. The calculation method is detailed in Segall et al. (1996), and the results are provided in Table S4 and shown in Figure 10. The lower the value of formation energy, the easier the substitution occurs, and a negative value of formation energy indicates that the substitution process can take place spontaneously (e.g., Xiao et al., 2023). In Jobs 3 and 4, the formation energies are –1.52 eV and –1.67 eV, respectively (Figs. 10A and 10B). These negative values suggest that both sphalerite and wurtzite can be readily incorporated with Ge via the substitution scheme of $\text{Ge}^{2+} \leftrightarrow \text{Zn}^{2+}$. In contrast, the formation energies are all positive values when Ge is incorporated via the substitution scheme of $\text{Ge}^{4+} + \text{vacancy} \leftrightarrow$

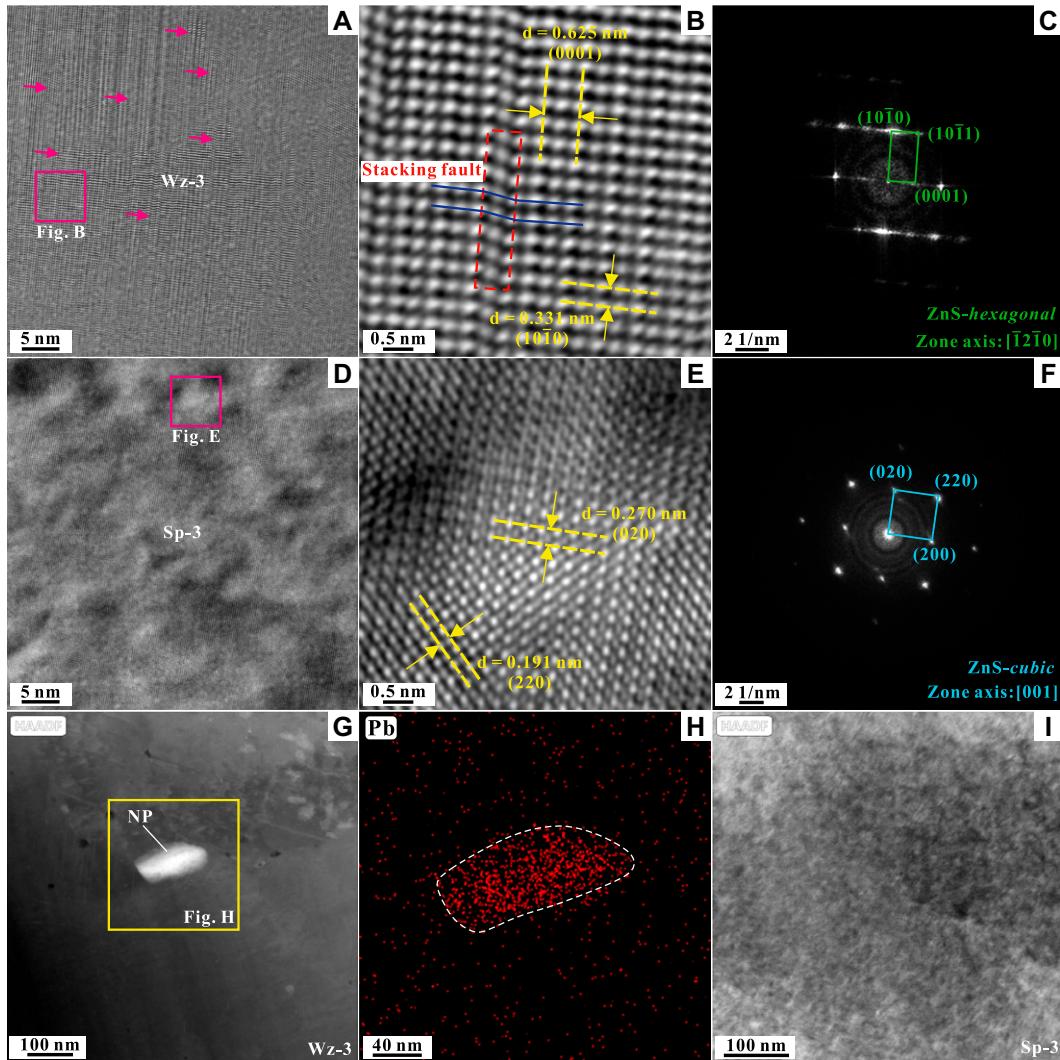


Figure 5. (A) High-resolution transmission electron microscopy (HRTEM) image showing stacking faults (pink arrows) in Wz-3. (B) Enlarged view of the region marked in part A, displaying the stacking fault and measured d-spacings in Wz-3. (C) Fast Fourier transform (FFT) image of the area in part B, revealing a hexagonal structure in Wz-3. (D) HRTEM image of Sp-3. (E) Magnified image of the squared area in part D, showing measured d-spacings in Sp-3. (F) FFT image of the area in part E, revealing a cubic structure in Sp-3. (G) High-angle annular dark field-scanning transmission electron microscopy (HAADF-STEM) image showing a nanoparticle in Wz-3. (H) Energy dispersive spectroscopy elemental map in part G, showing Pb element concentration. (I) HAADF-STEM image of Sp-3. NP—nanoparticle; Sp—sphalerite; Wz—wurtzite.

TABLE 2. STRUCTURAL DATA FOR SPHALERITE AND WURTZITE-2H IN THE LITERATURE

Mineral	Formula	Symmetry	System	Ideal d-spacing (d)	Reference
Sphalerite	ZnS	Cubic	F-43m	d(220) = 1.91 Å; d(020) = 2.705 Å	Skinner (1961)
Wurtzite-2H	ZnS	Hexagonal	P6 ₃ mc	d(0001) = 6.256 Å	Cui et al. (1992)

2Zn^{2+} (Jobs 5–8), regardless of whether it is in sphalerite or wurtzite. In Jobs 5 and 7, the incorporations of Ge^{4+} into sphalerite and wurtzite

were achieved by removing two adjacent Zn^{2+} and placing Ge^{4+} between them, resulting in formation energy values of 1.37 eV and 1.27 eV,

respectively (Figs. 10C and 10E). In Jobs 6 and 8, the incorporations of Ge^{4+} into sphalerite and wurtzite generated formation energy values of 1.83 eV and 1.74 eV, respectively, via substituting on a Zn^{2+} site and removing a distant Zn^{2+} (Figs. 10D and 10F).

In Jobs 3 and 4, Ge-S bond distances in sphalerite are consistently 2.59 Å (Fig. 10A), whereas Ge-S bond distances in wurtzite range from 2.56 Å to 2.57 Å (Fig. 10B). These changes of Ge-S bond distances reflect the large ionic size of Ge^{2+} (0.73 Å; Shannon, 1976) compared to Zn^{2+} (0.6 Å). In Jobs 5 and 7, Ge-S bond distances in sphalerite and wurtzite range from 2.22 Å to 2.30 Å (Fig. 10C) and 2.22 Å to 2.29 Å (Fig. 10E), respectively. In Jobs 6 and 8, Ge-S bond distances in sphalerite and wurtzite vary from 2.27 Å to 2.29 Å (Fig. 10D) and 2.26 Å to 2.29 Å (Fig. 10F), respectively. These shorter Ge-S bonds could be attributed to the smaller ionic radius of Ge^{4+} (0.39 Å; Shannon, 1976) compared to Ge^{2+} . Overall, when Ge^{2+} or Ge^{4+}

Mineral	Stage I	Stage II	Stage III
Sphalerite (Sp)	Sp-1	Sp-2a Sp-2b	Sp-3
Wurtzite (Wz)			Wz-3
Pyrite			
Galena			
Calcite			

Figure 6. Paragenetic sequence of the Qingshan deposit.

TABLE 3. SUMMARY OF LASER ABLATION-INDUCTIVELY COUPLED PLASMA-MASS SPECTROMETRY TRACE ELEMENT CONTENTS IN SPHALERITE AND WURTZITE FROM THE QINGSHAN DEPOSIT (IN PPM)

Element	Mn	Fe	Cu	Ga	Ge	As	Ag	Cd	In	Sn	Sb	Tl	Pb
Sphalerite-1 (n = 19)													
Maximum	137	66,736	27.0	0.51	672	761	12.1	2557	b.d.l.	0.16	8.16	3.60	48.8
Minimum	39.1	29,646	b.d.l.	b.d.l.	109	b.d.l.	0.20	324	b.d.l.	b.d.l.	b.d.l.	0.27	0.76
Mean	71.2	42,288	5.62		381		3.06	1078			2.55	1.41	14.7
Median	67.4	41,130	3.31		399		1.27	761			1.14	1.29	11.0
Sphalerite-2a (n = 23)													
Maximum	431	63,073	76.6	52.4	899	26.7	26.7	3253	0.02	3.00	19.5	8.42	158
Minimum	14.7	8917	b.d.l.	b.d.l.	174	b.d.l.	0.12	309	b.d.l.	b.d.l.	b.d.l.	0.25	1.13
Mean	113	32,315	16.4	3.26	435	5.22	7.53	1600			2.48	3.27	43.6
Median	86.3	27,806	5.35	0.45	333	1.97	5.53	1458			0.50	3.85	21.6
Sphalerite-2b (n = 23)													
Maximum	8.03	11,286	97.7	77.4	14.8	2.16	4.31	6779	0.48	43.7	3.94	0.09	7.90
Minimum	1.15	1720	3.37	0.63	b.d.l.	b.d.l.	0.09	883	b.d.l.	b.d.l.	b.d.l.	b.d.l.	0.04
Mean	2.87	3434	33.7	25.2	6.05		1.20	3288	0.05	4.76	0.60		1.02
Median	2.91	2740	22.8	20.3	4.53		0.80	3558	0.03	1.34	0.14		0.34
Wurtzite-3 (n = 41)													
Maximum	173	56,159	1400	42.7	722	3093	2034	13,888	0.06	3.74	2042	90.9	8892
Minimum	21.0	7347	18.4	b.d.l.	58.0	276	13.9	2872	b.d.l.	b.d.l.	7.24	1.26	203
Mean	63.5	22,379	577	13.0	253	712	436	5810		0.71	294	18.8	2187
Median	54.2	19,705	542	10.3	211	428	132	4576		0.38	113	10.1	1414
Sphalerite-3 (n = 34)													
Maximum	109	35,323	48.3	26.4	1393	377	77.6	7702	0.19	8.44	117	10.5	469
Minimum	19.4	9544	b.d.l.	b.d.l.	467	3.20	0.30	1901	b.d.l.	b.d.l.	1.28	1.25	75.7
Mean	46.9	18,081	12.1	1.94	840	18.2	15.7	4622		0.45	26.0	5.46	219
Median	44.0	16,668	4.63	0.33	827	16.9	7.03	4335		0.13	18.5	5.39	208

Note: b.d.l. denotes below detection limit; values below detection limit were replaced with their respective detection limits for the evaluation of mean and median values; elements with more than 50% of their values below detection limit were omitted from the calculation.

is introduced into the ZnS lattice, a slight distortion occurs in both sphalerite and wurtzite crystal structures (Table S5).

DISCUSSION

Incorporation Mechanisms of Ge in Sphalerite and Wurtzite

The presence of nano-to-micron-sized Ge-minerals in sphalerite has been reported from various sulfide deposits, such as Kipushi-type deposits (Melcher et al., 2006; Horn et al., 2019), metamorphosed Mississippi Valley-type or vein-type deposits in the Pyrenees (Cugerone et al.,

2020, 2021), and carbonate-hosted deposits in the Sichuan-Yunnan-Guizhou region (Meng et al., 2023; Sun et al., 2024; Niu et al., 2024). However, no micrometer-sized Ge mineral inclusions were found in our time-resolved LA-ICP-MS spectra and LA-ICP-MS mapping (Figs. 8 and 9). Two focused ion beam foil samples, obtained from positions close to the highest Ge contents (722 ppm in wurtzite and 1393 ppm in sphalerite) as determined by LA-ICP-MS measurements, were analyzed by TEM. The results revealed no nanoscale Ge-rich inclusions. The only nanoscale inclusions identified were those of galena within wurtzite (Figs. 5G and 5H). These observations suggest that Ge is typically

lattice-bound in both sphalerite and wurtzite at Qingshan.

Germanium is commonly positively correlated to monovalent cations such as Cu⁺ or Ag⁺, which suggests coupled substitution with Zn (e.g., Belissont et al., 2014, 2016; Bauer et al., 2019; Niu et al., 2024; Fougereuse et al., 2023). Nonetheless, these substitution mechanisms are not applicable to Cu-Ag-depleted but Ge-rich sphalerite at Qingshan (Fig. 7). Although wurtzite at Qingshan is enriched in Cu, Ag, and Ge, there is no positive correlation between Ge and Cu or Ag (Figs. 9, 11A, and 11B). Previous studies have also suggested the coupling of Ge with divalent cations (e.g., Mn²⁺, Fe²⁺,

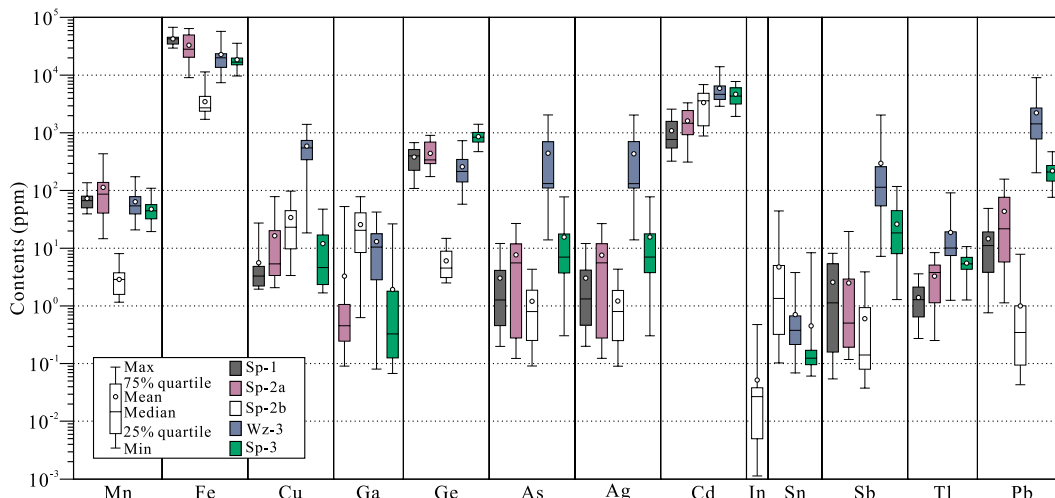


Figure 7. Box plot of laser ablation-inductively coupled plasma-mass spectrometry trace element contents in sphalerite (Sp) and wurtzite (Wz) from the Qingshan deposit.

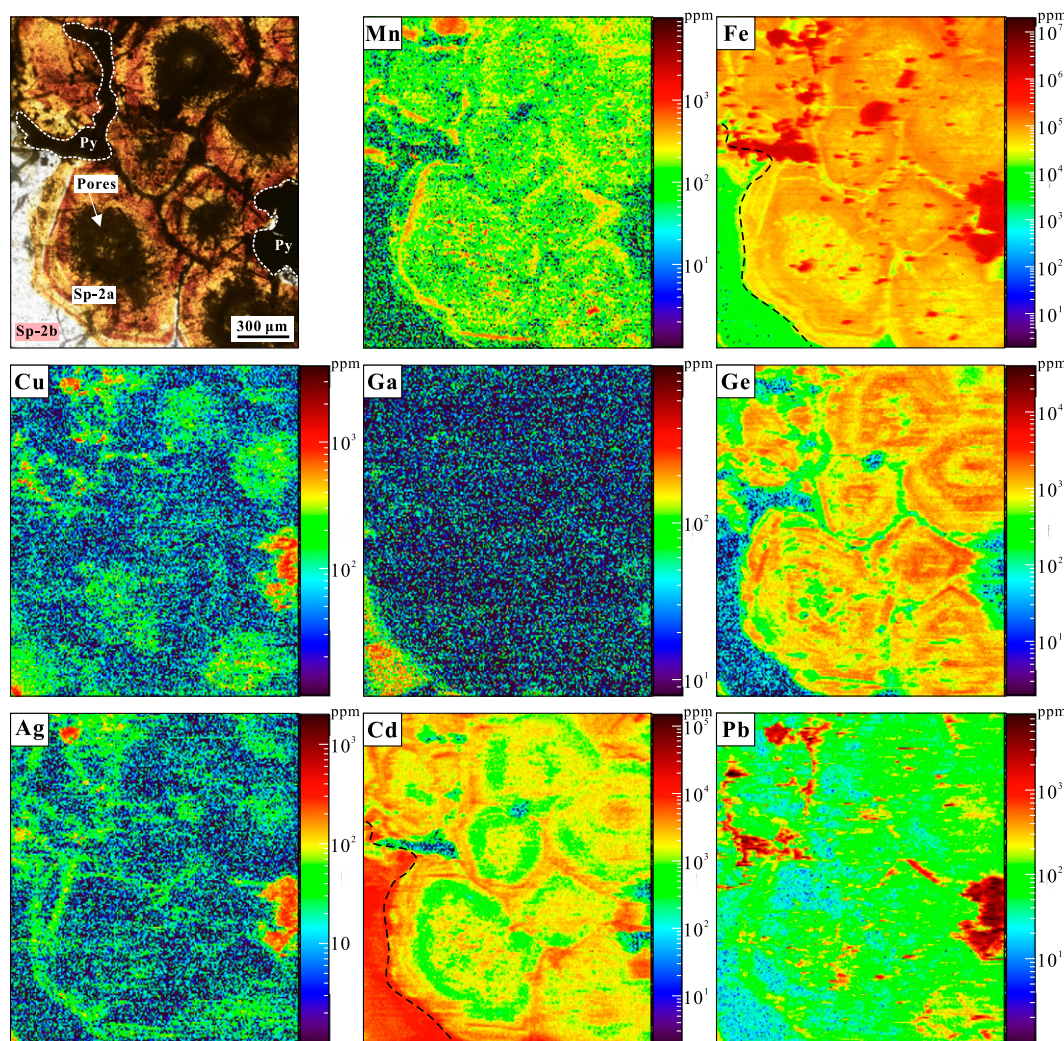


Figure 8. Laser ablation-inductively coupled plasma-mass spectrometry element mapping images of Sp-2a and Sp-2b. Py—pyrite; Sp—sphalerite.

Cd^{2+} , or Pb^{2+}) to substitute for Zn (e.g., Cook et al., 2009; Yuan et al., 2018; Hu et al., 2021; Luo et al., 2022; Bai et al., 2024). The elemental mapping shows that Ge contents seem broadly positively correlated with Fe in Sp-2a and Wz-3, as well as with Fe, Cd, and Pb in Sp-3 (Figs. 8 and 9). Binary correlation plots, however, reveal no positive correlations between Ge and these cations (Figs. 11C–11H).

Given that all common coupled substitutions mentioned above are ruled out, we propose that Ge in sphalerite and wurtzite at Qingshan directly substitutes for Zn. As highlighted by Liu et al. (2023), Ge can incorporate into the sphalerite lattice regardless of whether coupled substitution occurs. Although Ge in sphalerite is generally considered to be in the tetravalent oxidation state (Ge^{4+}), occasional divalent Ge (Ge^{2+}) has also been reported (Bonnet et al., 2017; White et al., 2022). No X-ray absorption near edge spectroscopy data are yet available for Ge-rich sphalerite and wurtzite at Qingshan.

Considering the two oxidation states of Ge (Ge^{2+} and Ge^{4+}), there are two possible direct substitution mechanisms for the incorporation of Ge into the sphalerite and wurtzite: $\text{Ge}^{2+} \leftrightarrow \text{Zn}^{2+}$ and $\text{Ge}^{4+} + \text{vacancy} \leftrightarrow 2\text{Zn}^{2+}$. Such direct substitution mechanisms have also been reported in other Zn-Pb deposits (e.g., Cook et al., 2015; Li et al., 2023; Liu et al., 2025).

Influence of Crystal Structures on Ge Incorporation

Based on LA-ICP-MS analyses, Bonnet et al. (2016) demonstrated a relationship between Ge contents and ZnS crystal structures: hexagonal ZnS (wurtzite) is more enriched in Ge than cubic ZnS (sphalerite). In contrast, our LA-ICP-MS analyses reveal that sphalerite (Sp-1, Sp-2a, and Sp-3) displays significantly higher Ge contents than wurtzite (Wz-3; Fig. 7). The density functional theory-based simulations were employed to investigate the influence of the two ZnS crys-

tal structures on Ge incorporation into their lattices. This study considered the direct substitution mechanisms as suggested above, i.e., $\text{Ge}^{2+} \leftrightarrow \text{Zn}^{2+}$ and $\text{Ge}^{4+} + \text{vacancy} \leftrightarrow 2\text{Zn}^{2+}$.

The negative formation energies of Ge^{2+} -bearing sphalerite (Fig. 10A) and Ge^{2+} -bearing wurtzite (Fig. 10B) indicate that Ge^{2+} can be readily incorporated into both crystal structures. On the contrary, both Ge^{4+} -bearing sphalerite (Figs. 10C and 10D) and Ge^{4+} -bearing wurtzite (Figs. 10E and 10F) have positive formation energies, suggesting that the incorporation of Ge^{4+} into them is energetically unfavorable. According to the Goldschmidt rule, the substitution of Ge^{4+} for Zn^{2+} is unlikely because the difference between the ionic radii of Ge^{4+} (0.39 Å) and Zn^{2+} (0.6 Å) is >15%. However, it is widely accepted that Ge is incorporated into ZnS in the tetravalent oxidation state, in which substitution of Ge^{4+} for Zn^{2+} is charge-balanced by the creation of lattice vacancies (Cook et al., 2015; Belissont et al., 2016; Liu et al., 2023).

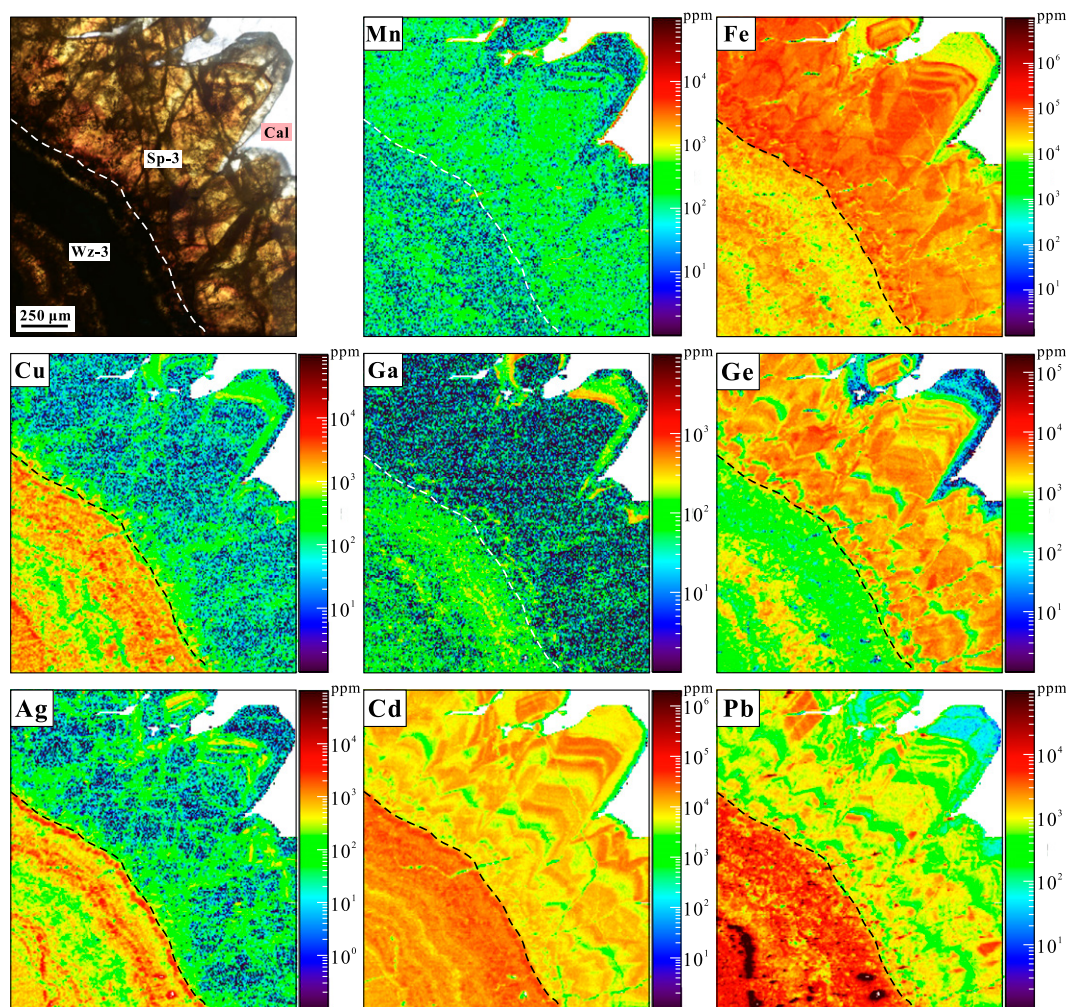


Figure 9. Laser ablation-inductively coupled plasma-mass spectrometry element mapping images of Wz-3 and Sp-3. Cal—calcite; Sp—sphalerite; Wz—wurtzite.

Overall, the similar formation energies of Ge-bearing sphalerite and Ge-bearing wurtzite under the same conditions (whether as Ge^{2+} or Ge^{4+} substitutions) suggest that the ability to incorporate Ge is comparable between sphalerite and wurtzite. In addition, the density functional theory-based simulations also show that Ge-bearing sphalerite and Ge-bearing wurtzite have similar variations in Ge-S bond distances when compared to the Zn-S bond distances in their respective standard crystals (Figs. 10A–10F; Table S5). This indicates that the two crystal structures of ZnS are both able to accommodate the same levels of Ge. Hence, the significant disparity in Ge contents between sphalerite and wurtzite should be attributed to physicochemical variations rather than their crystal structures.

Formation of Colloform Wurtzite

In some Zn-Pb sulfide deposits, the formation of colloform ZnS, particularly colloform sphalerite, is associated with (1) low temperature ($<100^\circ\text{C}$), (2) bacterial sulfate reduction,

and (3) supersaturated solutions (e.g., Barrie et al., 2009; Pfaff et al., 2011; Gagnevin et al., 2014; Luo et al., 2022; Mu et al., 2023). Commonly, bacterial sulfate reduction occurs at temperatures below 100°C (Jørgensen, 1979) and yields negative $\delta^{34}\text{S}$ values ($\Delta^{34}\text{S} = 40\text{‰}$ – 60‰ ; Machel, 2001; Basuki et al., 2008). However, sulfides at Qingshan, including colloform wurtzite, all have positive $\delta^{34}\text{S}$ values from $+10.7\text{‰}$ to $+19.6\text{‰}$ (Fig. 12), which are significantly higher than those of mantle-derived sulfur and exhibit the signature of thermochemical sulfate reduction, i.e., enrichment in heavy sulfur isotopes (Ohmoto, 1986). Generally, the triggering temperature of thermochemical sulfate reduction is at least 120°C (Machel et al., 1995; Seal, 2006). Using the mineral geothermometer GGIMFis (Ga, Ge, In, Mn, and Fe in sphalerite) proposed by Frenzel et al. (2016), we obtained formation temperatures of 82 – 183°C (median 121°C) for Wz-3 (Table S8). Therefore, it can be concluded that low temperatures ($<100^\circ\text{C}$) and bacterial sulfate reduction are not necessarily correlated with colloform tex-

tures. Considering the absence of sulfate at Qingshan, average $\delta^{34}\text{S}$ value of the sulfides ($+16.8\text{‰}$) could represent the total sulfur isotopic composition of the fluid (e.g., Ohmoto, 1972). This value is similar to $\delta^{34}\text{S}$ values of Carboniferous seawater (Fig. 12).

Crystal structural defects can be attributed to rapid crystallization from supersaturated solutions (Minčeva-Stefanova, 1993). The abundant stacking faults in Wz-3 (Figs. 5A and 5B) imply that the formation of its colloform texture may be associated with supersaturated solutions. This interpretation is further supported by the presence of micron- to nanometer-sized galena inclusions in Wz-3 (Figs. 4G and 5G), as Pb can reach high concentrations in colloform texture due to sudden oversaturation (Roedder, 1968; Pfaff et al., 2011). Fluid mixing may be the most common means by which the supersaturation required for colloform textures is achieved in nature (Gagnevin et al., 2014), though it cannot account for all colloform minerals.

Wurtzite is generally regarded as the high-temperature ZnS phase (Subramani et al.,

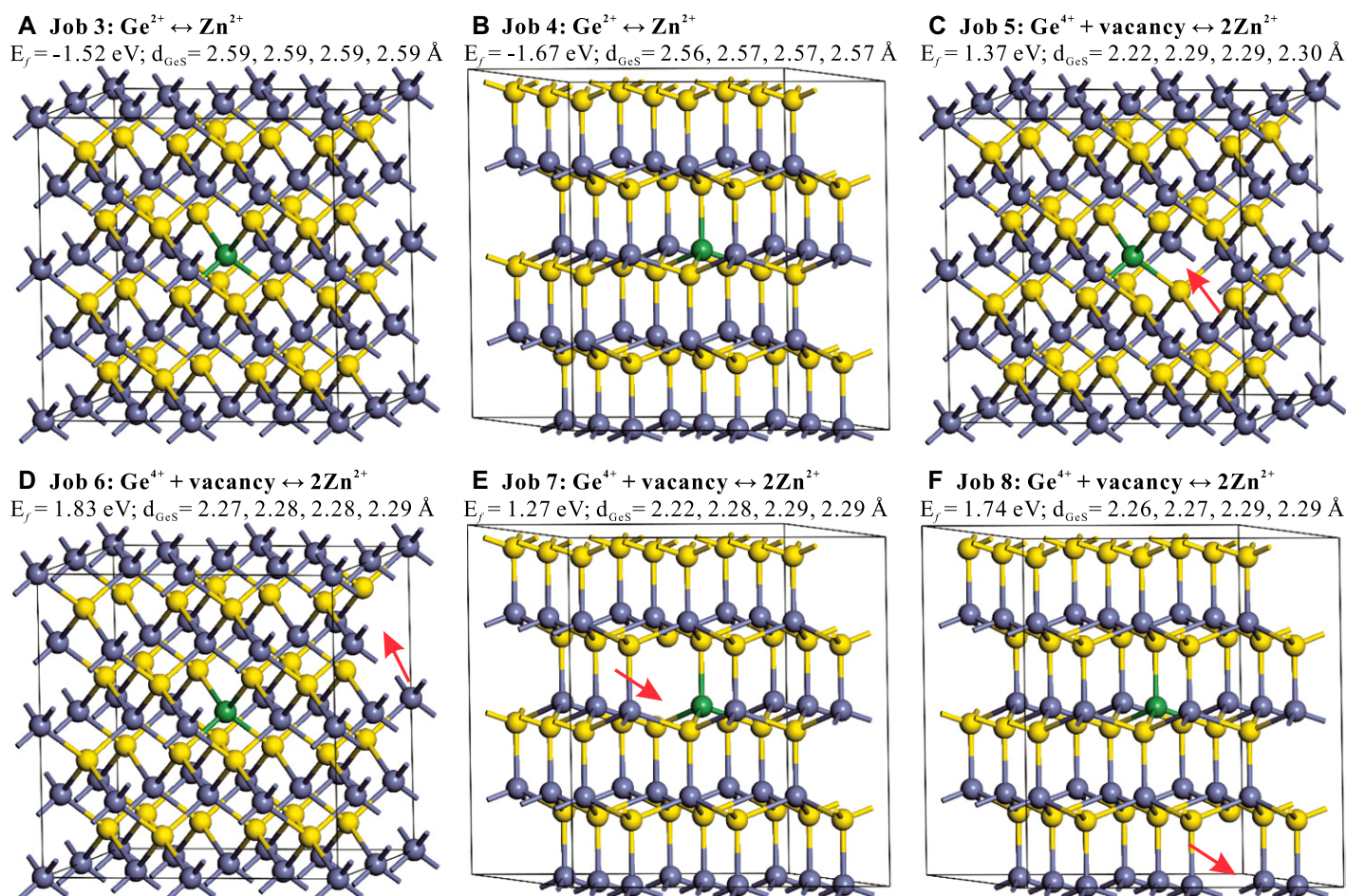


Figure 10. Systems of optimized geometry for Ge incorporation in ZnS crystal structures. (A) Ge incorporation in sphalerite via $\text{Ge}^{2+} \leftrightarrow \text{Zn}^{2+}$. (B) Ge incorporation in wurtzite via $\text{Ge}^{2+} \leftrightarrow \text{Zn}^{2+}$. (C, D) Ge incorporation in sphalerite via $\text{Ge}^{4+} + \text{vacancy} \leftrightarrow 2\text{Zn}^{2+}$. (E, F) Ge incorporation in wurtzite via $\text{Ge}^{4+} + \text{vacancy} \leftrightarrow 2\text{Zn}^{2+}$. Blue-gray ball—Zn; yellow ball—S; green ball—Ge; E_f —formation energy; d_{GeS} —Ge-S bond distance; red arrow points vacancy.

2023). However, wurtzite and sphalerite can coexist in some Mississippi Valley-type Zn-Pb deposits (e.g., Bonnet et al., 2016; Torró et al., 2023), which typically form at low temperatures (90–150 °C; Leach et al., 2005). Actually, Scott and Barnes (1972) showed by hydrothermal experiments that wurtzite-sphalerite equilibrium is a univariant function of sulfur fugacity over a wide temperature range, and wurtzite is thermodynamically stable at lower sulfur fugacity than sphalerite. In other words, whether ZnS crystallizes in the wurtzite or sphalerite form is constrained by sulfur fugacity. Additionally, it has been suggested that the temperature of inversion between sphalerite and wurtzite can be lowered by the presence of trace elements (Ueno et al., 1996; Knitter and Binnewies, 1999; Subramani et al., 2023). These studies demonstrate that solid solutions in ZnS (e.g., Fe, Mn, and Cd) shift the sphalerite-wurtzite phase boundary toward higher sulfur fugacity, thereby

lowering the transition temperature from sphalerite to wurtzite. This shift essentially involves the incorporation of exotic elements, which is accommodated by modifications to the crystal structure (Xu et al., 2021). In our samples, Wz-3 is more enriched in Mn, Fe, and Cd compared to its closely associated Sp-3 (Fig. 7).

In summary, supersaturation-driven rapid crystallization and low sulfur fugacity, combined with the enrichment of Mn, Fe, and Cd, could therefore explain the existence of colloform wurtzite at Qingshan.

Implications for Ge Enrichment Processes

Coupled dissolution-precipitation plays a crucial role in numerous geological processes by enabling the remobilization and re-enrichment of elements (Putnis, 2002; Altree-Williams et al., 2015). Core-rim textures observed in our samples are attributed to coupled dissolution-

reprecipitation induced by multi-pulse fluids, as evidenced by sharp boundaries, porosity, and heterogeneous textures (Figs. 4D–4F). These features can serve as diagnostic criteria for coupled dissolution-reprecipitation (Putnis, 2002, 2009; Li et al., 2021; Ehrig et al., 2023; Yang et al., 2024). The heterogeneous Ge distribution in sphalerite (i.e., Ge-enriched Sp-2a versus Ge-depleted Sp-2b domains; Fig. 7) may indicate an interaction between fluids and sphalerite, as previously suggested by Cugerone et al. (2021). During fluid-mediated coupled dissolution-reprecipitation, Ge was likely released from the crystal lattice of parent Sp-2 and subsequently concentrated in Wz-3 and Sp-3, leaving behind Ge-depleted Sp-2b ($\leq 14.8 \text{ ppm}$). This process is schematically illustrated in Figure 13. Here, additional Ge from the external fluid source could not be completely ruled out. Although both phases are Ge-enriched, euhedral Sp-3 (467–1393 ppm; median 840 ppm) exhibits sig-

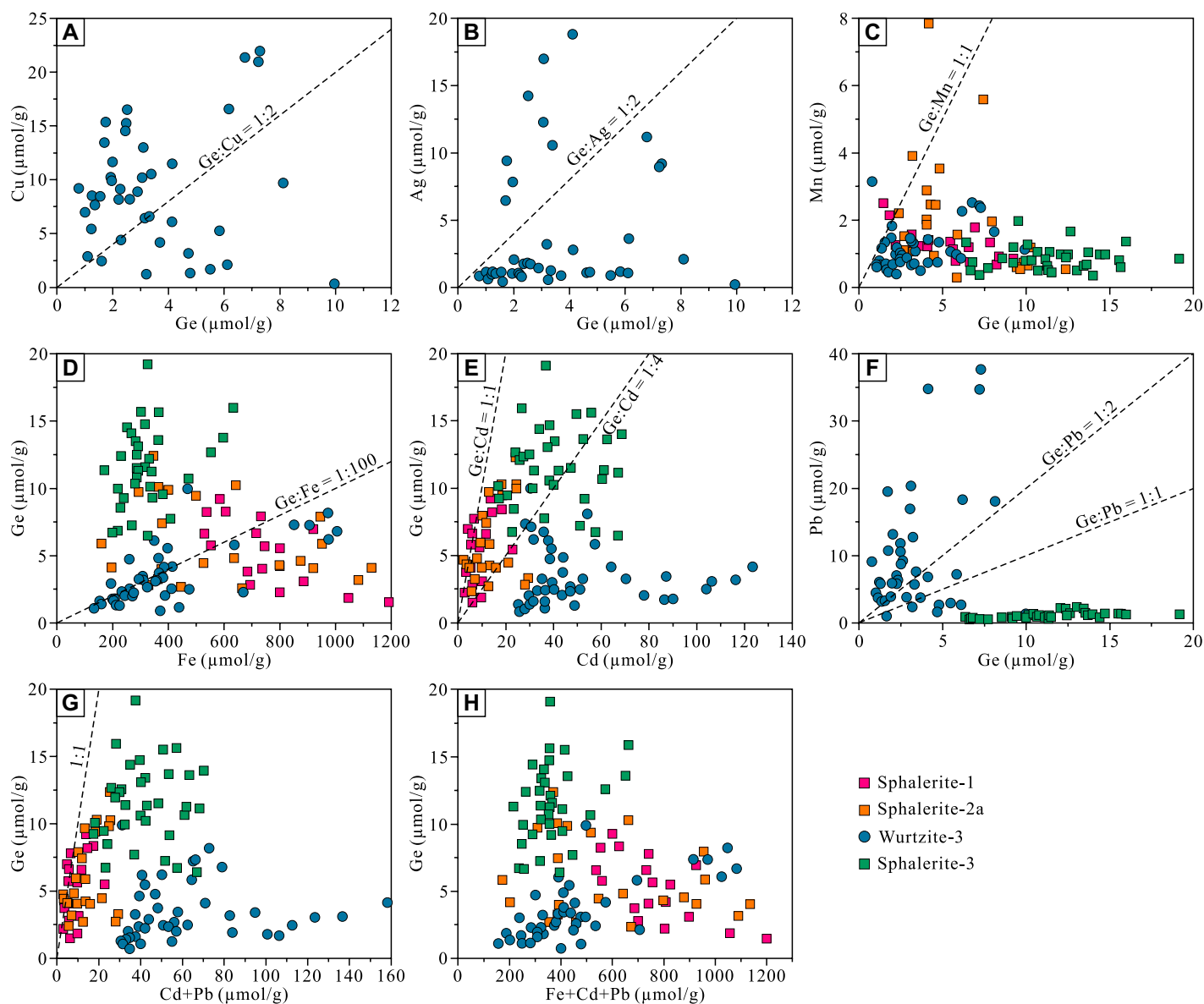


Figure 11. Binary correlation plots of laser ablation–inductively coupled plasma–mass spectrometry trace elements of sphalerite and wurtzite from the Qingshan deposit.

nificantly higher Ge contents compared to colloform Wz-3 (58.0–722 ppm; median 211 ppm). The enhanced incorporation of Ge into Sp-3 seems to be driven by increasing sulfur fugacity rather than changing crystal structure, as the transition between wurtzite and sphalerite is dependent on sulfur fugacity.

Germanium redistribution has also been observed in other ore deposits. For example, the Arre vein-type sulfide deposit shows Ge remobilization from sphalerite and subsequent concentration in neocrystallized Ge minerals via fluid-assisted solution transfer during dynamic recrystallization of sphalerite (Cugerone et al., 2020). Determining how Ge concentrates at

the mineral scale is essential to understand the spatial distribution and localization at the deposit scale.

CONCLUSIONS

Three mineralization stages with five types of ZnS minerals (Sp-1, Sp-2a, Sp-2b, Wz-3, and Sp-3) are identified in the Qingshan deposit. All types exhibit Ge enrichment (up to 1393 ppm), except for Sp-2b, which is Ge-depleted (≤ 14.8 ppm). TEM analyses reveal that the incorporation of Ge in these ZnS minerals is lattice-bound. Two direct substitution mechanisms are proposed, depending on the

two possible Ge oxidation states: $\text{Ge}^{2+} \leftrightarrow \text{Zn}^{2+}$ and $\text{Ge}^{4+} + \text{vacancy} \leftrightarrow 2\text{Zn}^{2+}$. Based on density functional theory calculations, both ZnS crystal structures (cubic sphalerite and hexagonal wurtzite) can incorporate and accommodate Ge at similar levels, and the substitution scheme $\text{Ge}^{2+} \leftrightarrow \text{Zn}^{2+}$ is energetically and structurally more favorable.

The colloform texture of Wz-3 does not necessarily correlate with low temperatures ($< 100^\circ\text{C}$) and bacterial sulfate reduction. The precipitation of colloform Wz-3 was controlled by a combination of highly supersaturated solutions, low sulfur fugacity, and enrichments of Mn, Fe, and Cd. The increase in sulfur fugacity likely triggered

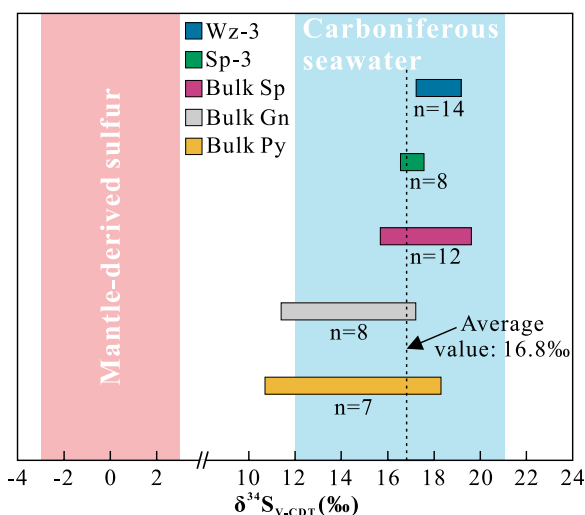


Figure 12. Sulfur isotopic compositions of sulfides from the Qingshan deposit. Bulk sulfide minerals at Qingshan were collected from Zhou et al. (2013); the Carboniferous seawater and mantle-derived sulfide are based on Claypool et al. (1980) and Chaussidon et al. (1989), respectively. Gn—galena; Py—pyrite; Sp—sphalerite; Wz—wurtzite; V-CDT—Vienna Canyon Diablo Troilite.

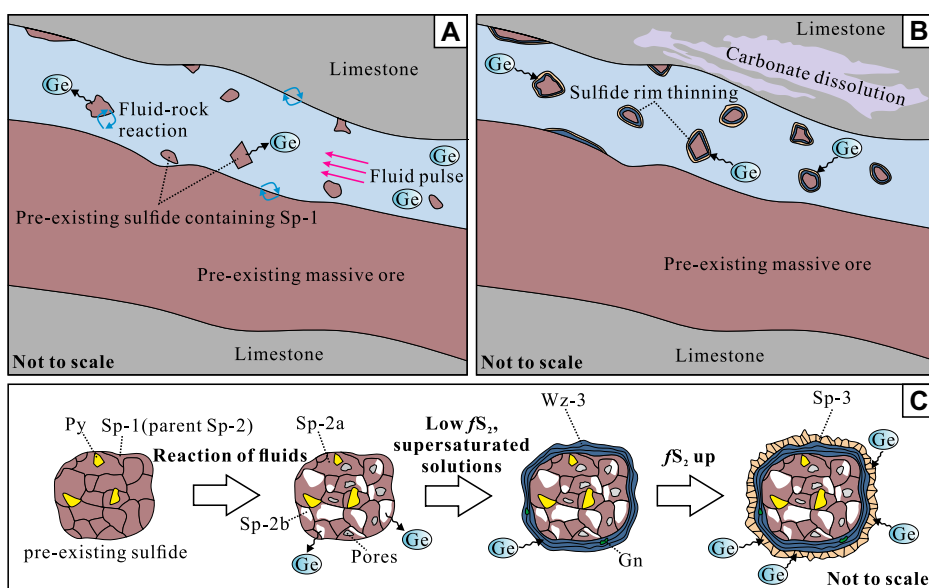


Figure 13. A cartoon showing the process of coupled dissolution-precipitation reactions responsible for Ge enrichment in the Qingshan deposit. (A) Ore fluid infiltration and fluid-rock reaction. Germanium was remobilized from preexisting sulfide into fluids. (B) Dissolution of carbonate and precipitation of sulfide growth rims. Germanium was incorporated into sulfide growth rims. (C) Detailed processes of sulfide growth and Ge enrichment associated with dissolution-precipitation reaction. The formation of Sp-3 is attributed to elevated fS_2 , which also promotes more significant Ge incorporation. Py—pyrite; Sp—sphalerite; Wz—wurtzite; fS_2 —sulfur fugacity.

the transition from colloform Wz-3 to euhedral Sp-3, resulting in higher Ge content in the latter (median 840 ppm) than in the former (median 211 ppm). Fluid-mediated coupled dissolution-precipitation likely induced Ge remobilization, leading to the development of Ge-depleted Sp-2b. Future X-ray absorption near edge spectroscopy analyses will be needed to determine the oxidation state of Ge in sphalerite and wurtzite from the Qingshan deposit.

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